Name: Sushant Balunda

Batch: B.Tech CSE (Core), JECRC University

Project Title: AWS VPC Peering (Project 01 / Oct 2026)

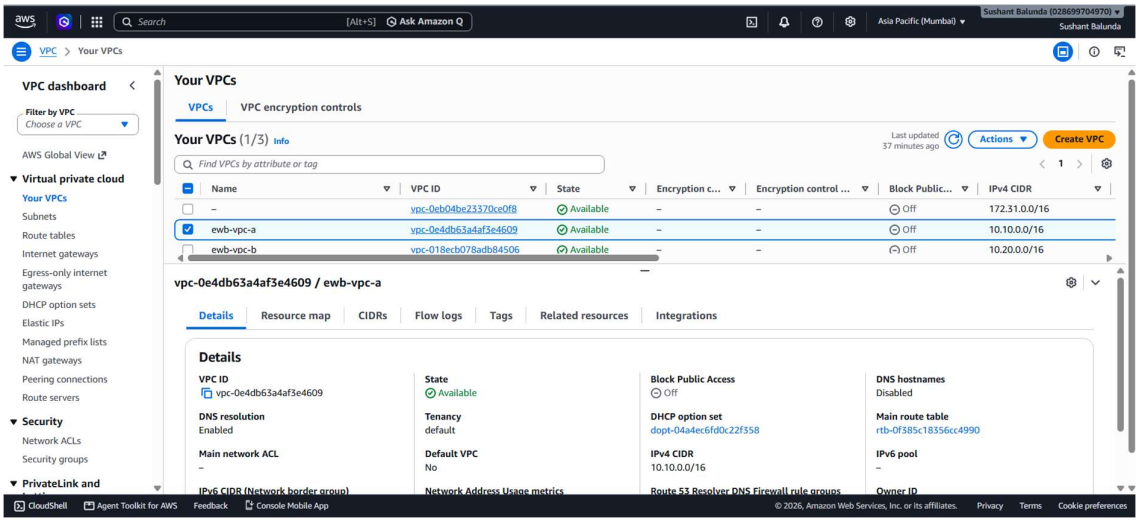
Region: Asia Pacific (Mumbai) — ap-south-1

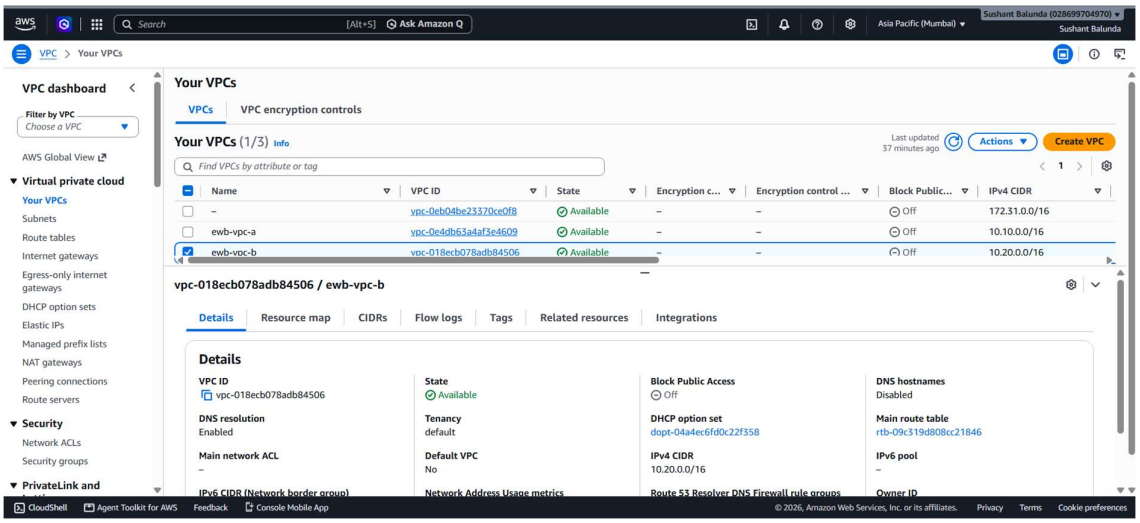
Date: October 20, 2026

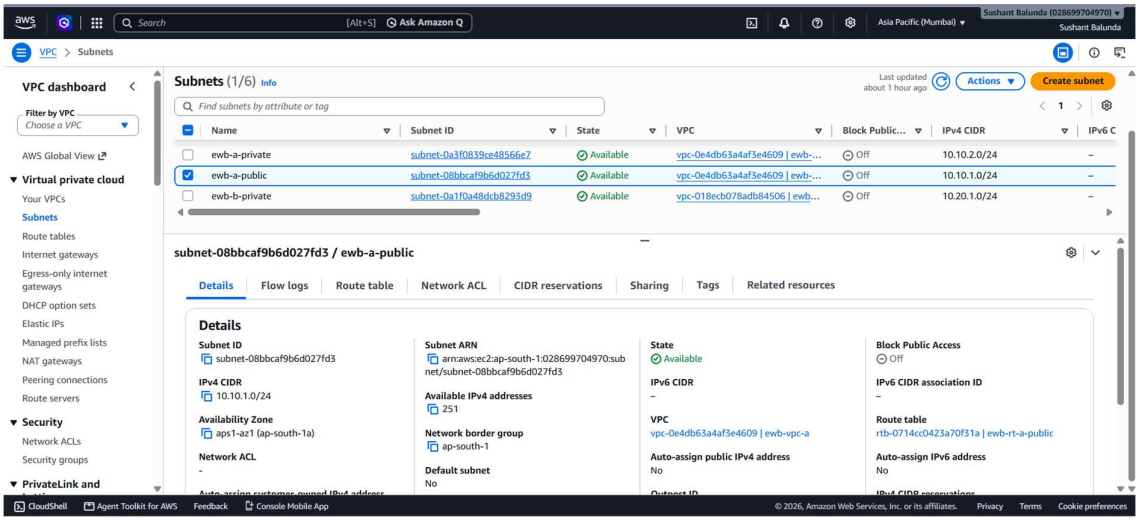
Section 1: Architecture Diagram and Resource Inventory

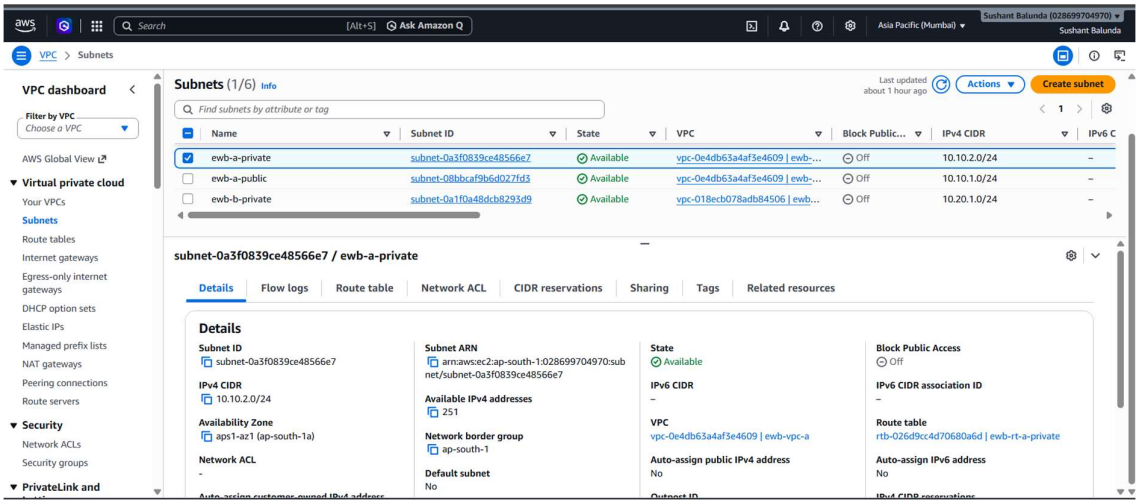
Resource Inventory:

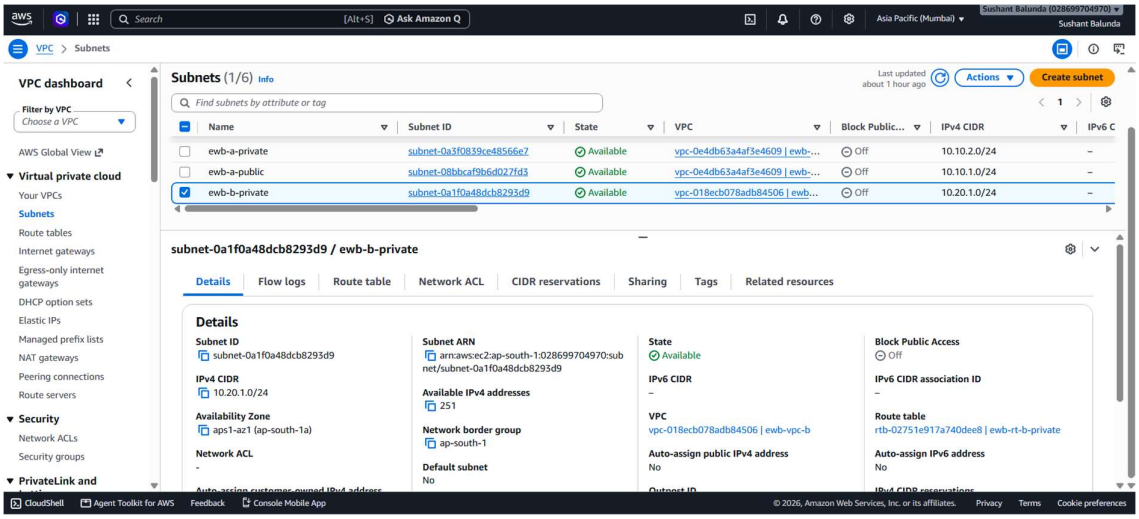
- **VPC A:** ewb-vpc-a with IPv4 CIDR 10.10.0.0/16
- **VPC A Public Subnet:** ewb-a-public with IPv4 CIDR 10.10.1.0/24
- **VPC A Private Subnet:** ewb-a-private with IPv4 CIDR 10.10.2.0/24
- **VPC B:** ewb-vpc-b with IPv4 CIDR 10.20.0.0/16
- **VPC B Private Subnet:** ewb-b-private with IPv4 CIDR 10.20.1.0/24
- **Bastion Host:** ewb-bastion located in ewb-a-public with Private IP 10.10.1.10 and Auto-assign Public IP enabled
- **App-A Instance:** ewb-app-a located in ewb-a-private with Private IP 10.10.2.10 and no Public IP
- **App-B Instance:** ewb-app-b located in ewb-b-private with Private IP 10.20.1.10 and no Public IP
- **Peering Connection:** ewb-a-b-peer (Active)

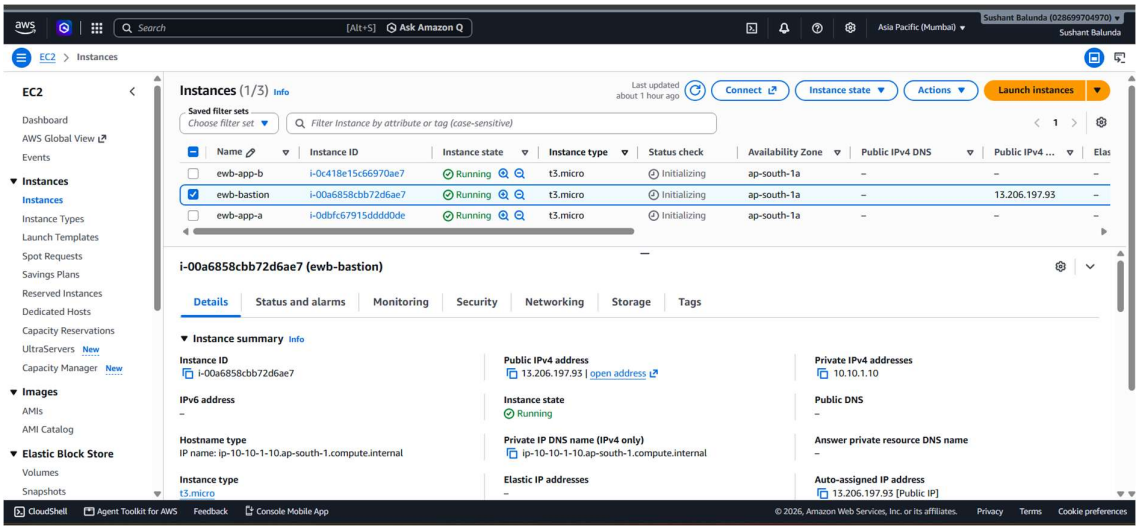


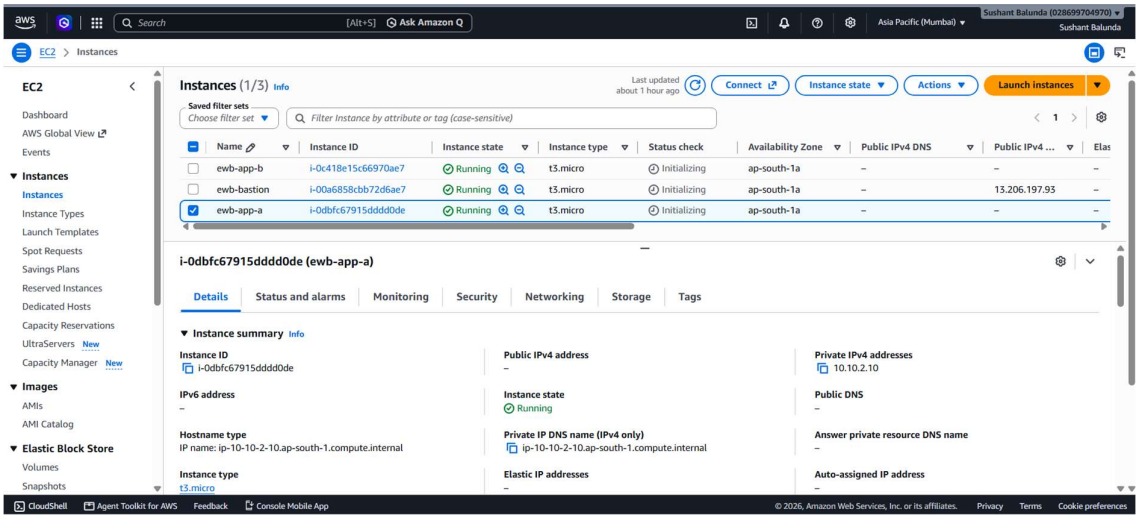

Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Your VPCs
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
Your VPCs
VPCs
VPC encryption controls
Your VPCs (1/3) Info
Last updated 37 minutes ago
Actions
Create VPC
Find VPCs by attribute or tag
Name
VPC ID
State
Encryption c...
Encryption control ...
Block Public...
IPv4 CIDR
– vpc-0eb04be23370ce0f8 Available – – Off 172.31.0.0/16
ewb-vpc-a vpc-0e4db63a4af3e4609 Available – – Off 10.10.0.0/16
ewb-vpc-b vpc-018ecb078adb84506 Available – – Off 10.20.0.0/16
vpc-018ecb078adb84506 / ewb-vpc-b
Details
Resource map
CIDRs
Flow logs
Tags
Related resources
Integrations
Details
VPC ID vpc-018ecb078adb84506
State Available
Block Public Access Off
DNS hostnames Disabled
DNS resolution Enabled
Tenancy default
DHCP option set dopt-04a4ec6fd0c22f358
Main route table rtb-09c319d808cc21846
Main network ACL –
Default VPC No
IPv4 CIDR 10.20.0.0/16
IPv6 pool –
IPv6 CIDR (Network border group)
Network Address Usage metrics
Route 53 Resolver DNS Firewall rule groups
Owner ID
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

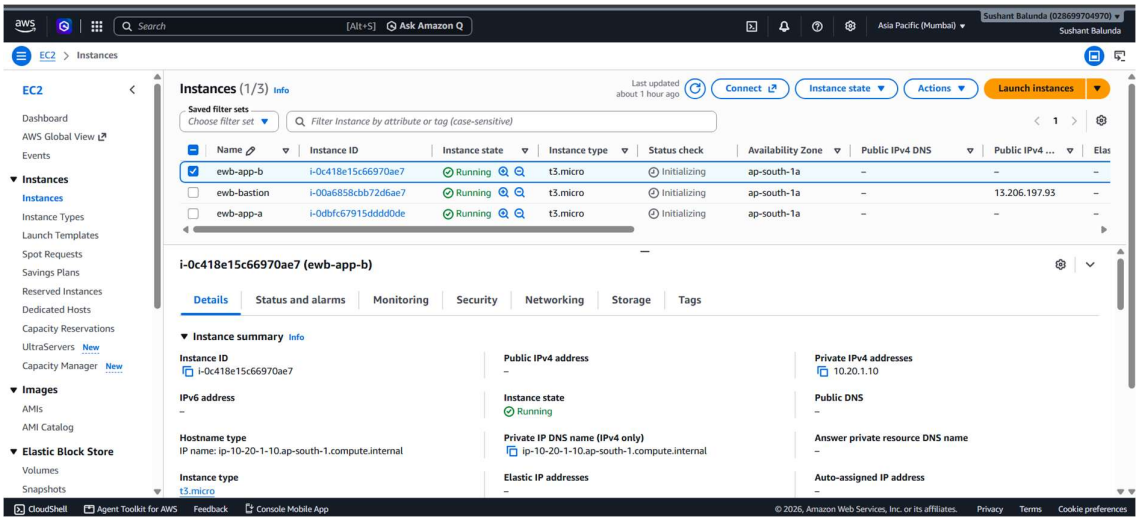

Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Subnets
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
Subnets (1/6) Info
Last updated about 1 hour ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
IPv6 C
ewb-a-private subnet-0a3f0839ce48566e7 Available vpc-0e4db63a4af3e4609 | ewb-... Off 10.10.2.0/24 –
ewb-a-public subnet-08bbcaf9b6d027fd3 Available vpc-0e4db63a4af3e4609 | ewb-... Off 10.10.1.0/24 –
ewb-b-private subnet-0a1f0a48dcb8293d9 Available vpc-018ecb078adb84506 | ewb... Off 10.20.1.0/24 –
subnet-08bbcaf9b6d027fd3 / ewb-a-public
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Details
Subnet ID subnet-08bbcaf9b6d027fd3
Subnet ARN arn:aws:ec2:ap-south-1:028699704970:subnet/subnet-08bbcaf9b6d027fd3
State Available
Block Public Access Off
IPv4 CIDR 10.10.1.0/24
Available IPv4 addresses 251
IPv6 CIDR –
IPv6 CIDR association ID –
Availability Zone aps1-az1 (ap-south-1a)
Network border group ap-south-1
VPC vpc-0e4db63a4af3e4609 | ewb-vpc-a
Route table rtb-0714cc0423a70f31a | ewb-rt-a-public
Network ACL –
Default subnet No
Auto-assign public IPv4 address No
Auto-assign IPv6 address No
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Subnets
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
Subnets (1/6) Info
Last updated about 1 hour ago
Actions
Create subnet
Find subnets by attribute or tag
Name
Subnet ID
State
VPC
Block Public...
IPv4 CIDR
IPv6 C
ewb-a-private subnet-0a3f0839ce48566e7 Available vpc-0e4db63a4af3e4609 | ewb-... Off 10.10.2.0/24 –
ewb-a-public subnet-08bbcaf9b6d027fd3 Available vpc-0e4db63a4af3e4609 | ewb-... Off 10.10.1.0/24 –
ewb-b-private subnet-0a1f0a48dcb8293d9 Available vpc-018ecb078adb84506 | ewb... Off 10.20.1.0/24 –
subnet-0a3f0839ce48566e7 / ewb-a-private
Details
Flow logs
Route table
Network ACL
CIDR reservations
Sharing
Tags
Related resources
Details
Subnet ID subnet-0a3f0839ce48566e7
Subnet ARN arn:aws:ec2:ap-south-1:028699704970:subnet/subnet-0a3f0839ce48566e7
State Available
Block Public Access Off
IPv4 CIDR 10.10.2.0/24
Available IPv4 addresses 251
IPv6 CIDR –
IPv6 CIDR association ID –
Availability Zone aps1-az1 (ap-south-1a)
Network border group ap-south-1
VPC vpc-0e4db63a4af3e4609 | ewb-vpc-a
Route table rtb-026d9cc4d70680a6d | ewb-rt-a-private
Network ACL –
Default subnet No
Auto-assign public IPv4 address No
Auto-assign IPv6 address No


VPC > Subnets
VPC dashboard
Subnets (1/6) Info
Create subnet
ewb-a-private subnet-0a3f0839ce48566e7 Available 10.10.2.0/24
ewb-a-public subnet-08bbcaf9b6d027fd3 Available 10.10.1.0/24
ewb-b-private subnet-0a1f0a48dcb8293d9 Available 10.20.1.0/24
subnet-0a1f0a48dcb8293d9 / ewb-b-private
Details
Subnet ID subnet-0a1f0a48dcb8293d9
IPv4 CIDR 10.20.1.0/24
Availability Zone aps1-az1 (ap-south-1a)
Subnet ARN arn:aws:ec2:ap-south-1:028699704970:subnet/subnet-0a1f0a48dcb8293d9
Available IPv4 addresses 251
Network border group ap-south-1
Default subnet No
State Available
IPv6 CIDR -
VPC vpc-018ecb078adb84506 | ewb-vpc-b
Auto-assign public IPv4 address No
Block Public Access Off
IPv6 CIDR association ID -
Route table rtb-02751e917a740dee8 | ewb-rt-b-private
Auto-assign IPv6 address No


EC2 > Instances
Instances (1/3) Info
Launch instances
ewb-app-b i-0c418e15c66970ae7 Running t3.micro Initializing ap-south-1a
ewb-bastion i-00a6858cbb72d6ae7 Running t3.micro Initializing ap-south-1a 13.206.197.93
ewb-app-a i-0dbfc67915dddd0de Running t3.micro Initializing ap-south-1a
i-00a6858cbb72d6ae7 (ewb-bastion)
Instance summary Info
Instance ID i-00a6858cbb72d6ae7
Public IPv4 address 13.206.197.93 | open address
Private IPv4 addresses 10.10.1.10
Instance state Running
Private IP DNS name (IPv4 only) ip-10-10-1-10.ap-south-1.compute.internal
Instance type t3.micro

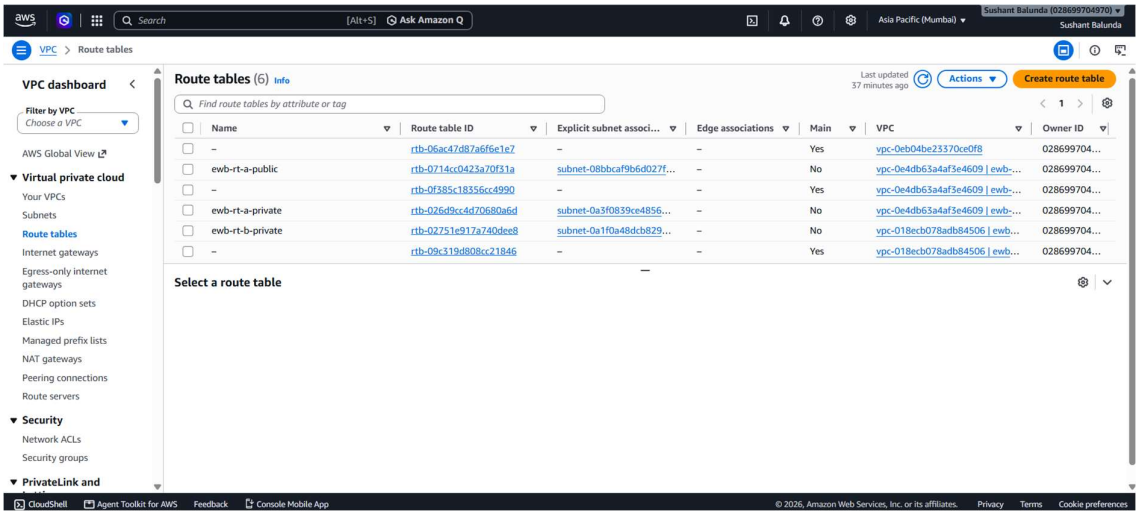

EC2 > Instances
Instances (1/3) Info
Launch instances
ewb-app-b i-0c418e15c66970ae7 Running t3.micro Initializing ap-south-1a
ewb-bastion i-00a6858cbb72d6ae7 Running t3.micro Initializing ap-south-1a 13.206.197.93
ewb-app-a i-0dbfc67915dddd0de Running t3.micro Initializing ap-south-1a
i-0dbfc67915dddd0de (ewb-app-a)
Instance summary Info
Instance ID i-0dbfc67915dddd0de
Public IPv4 address -
Private IPv4 addresses 10.10.2.10
Instance state Running
Private IP DNS name (IPv4 only) ip-10-10-2-10.ap-south-1.compute.internal
Instance type t3.micro

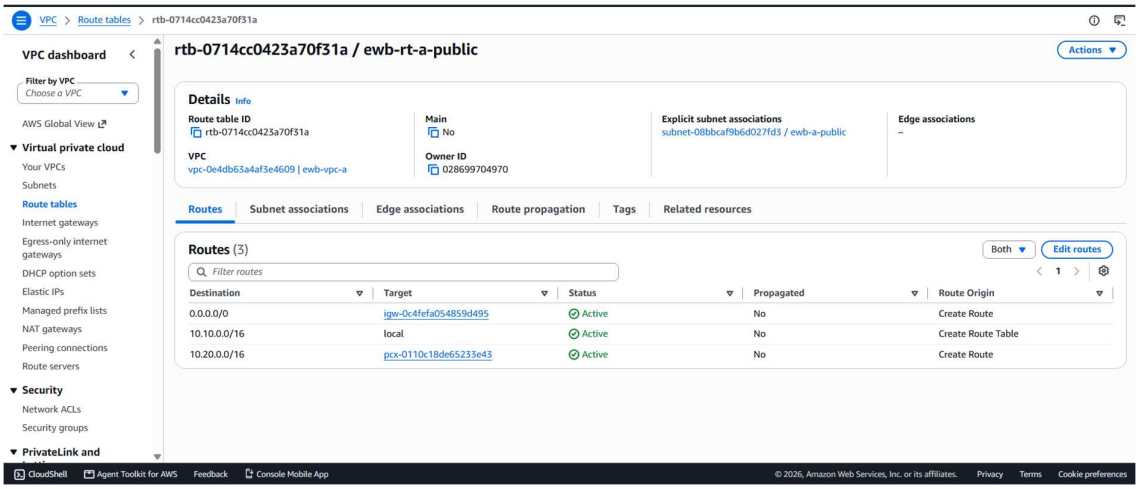

Search
[Alt+S]
Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
EC2 > Instances
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Instances (1/3) Info
Last updated about 1 hour ago
Connect
Instance state
Actions
Launch instances
Saved filter sets
Choose filter set
Filter Instance by attribute or tag (case-sensitive)
Name
Instance ID
Instance state
Instance type
Status check
Availability Zone
Public IPv4 DNS
Public IPv4 ...
ewb-app-b
i-0c418e15c66970ae7
Running
t3.micro
Initializing
ap-south-1a
ewb-bastion
i-00a6858cbb72d6ae7
Running
t3.micro
Initializing
ap-south-1a
13.206.197.93
ewb-app-a
i-0dbfc67915dddd0de
Running
t3.micro
Initializing
ap-south-1a
i-0c418e15c66970ae7 (ewb-app-b)
Details
Status and alarms
Monitoring
Security
Networking
Storage
Tags
Instance summary Info
Instance ID
i-0c418e15c66970ae7
Public IPv4 address
–
Private IPv4 addresses
10.20.1.10
IPv6 address
–
Instance state
Running
Public DNS
–
Hostname type
IP name: ip-10-20-1-10.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-10-20-1-10.ap-south-1.compute.internal
Answer private resource DNS name
–
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
–
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

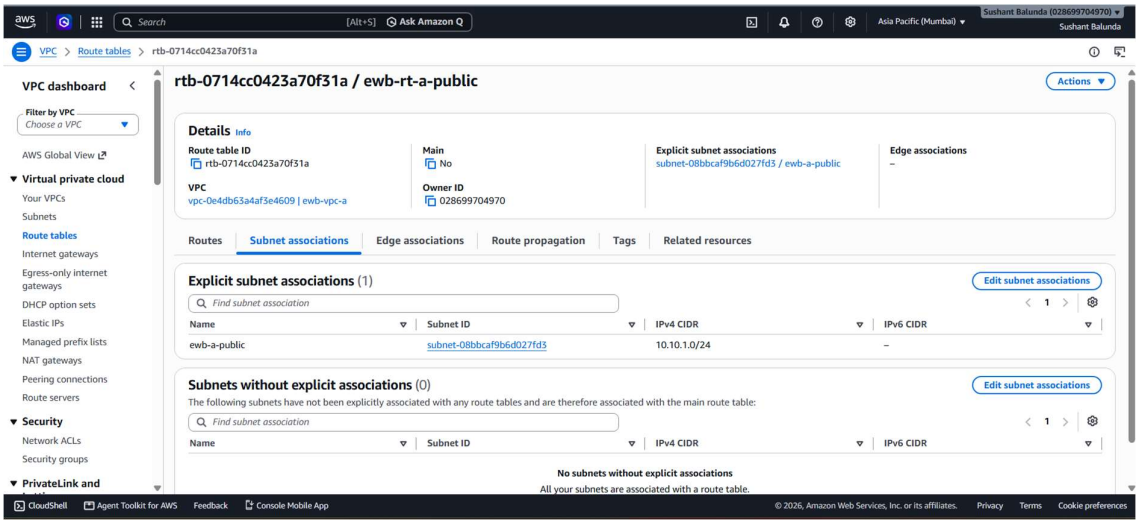
2. Route Tables and Subnet Associations

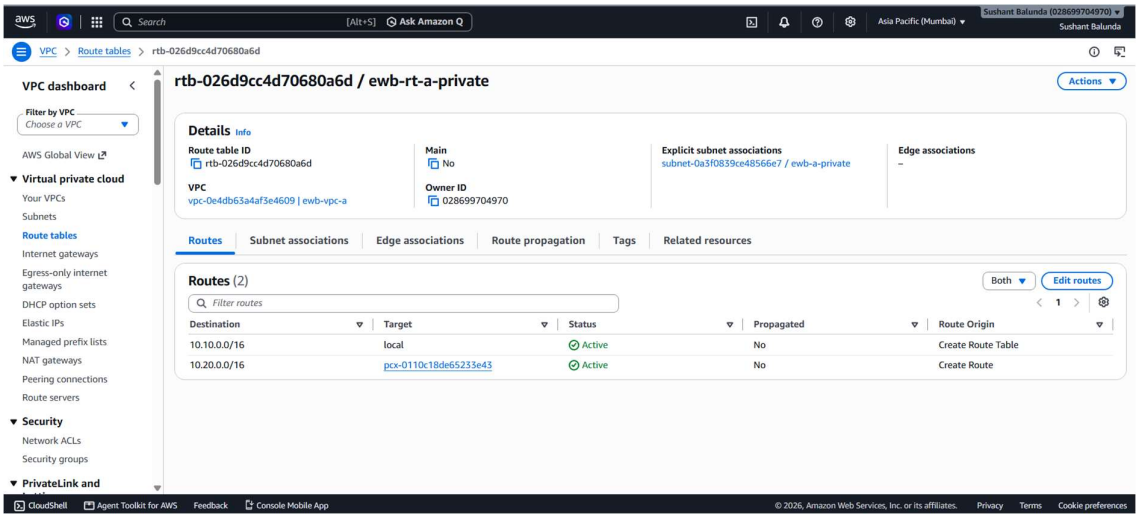
Routing Configuration:

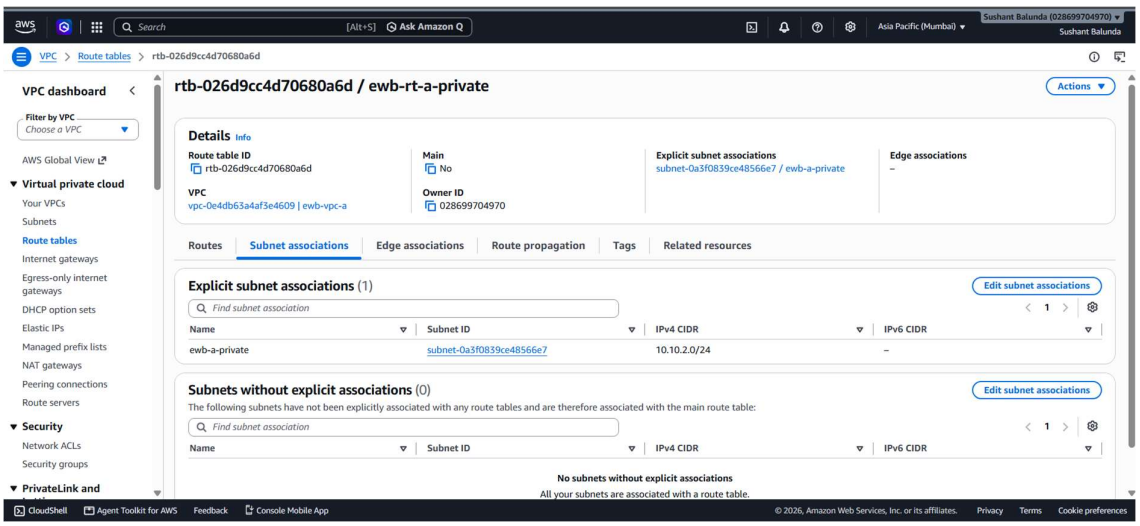
- **ewb-rt-a-public** (Associated with ewb-a-public): Contains a route to 0.0.0.0/0 via the Internet Gateway (ewb-igw-a) for bastion internet access, and a route to 10.20.0.0/16 via the peering connection (pcx-...) for bastion SSH access to App-B.
- **ewb-rt-a-private** (Associated with ewb-a-private): Contains a route to 10.20.0.0/16 via the peering connection (pcx-...) to route traffic from App-A to VPC B.
- **ewb-rt-b-private** (Associated with ewb-b-private): Contains a return route to 10.10.0.0/16 via the peering connection (pcx-...) to route traffic back to VPC A.

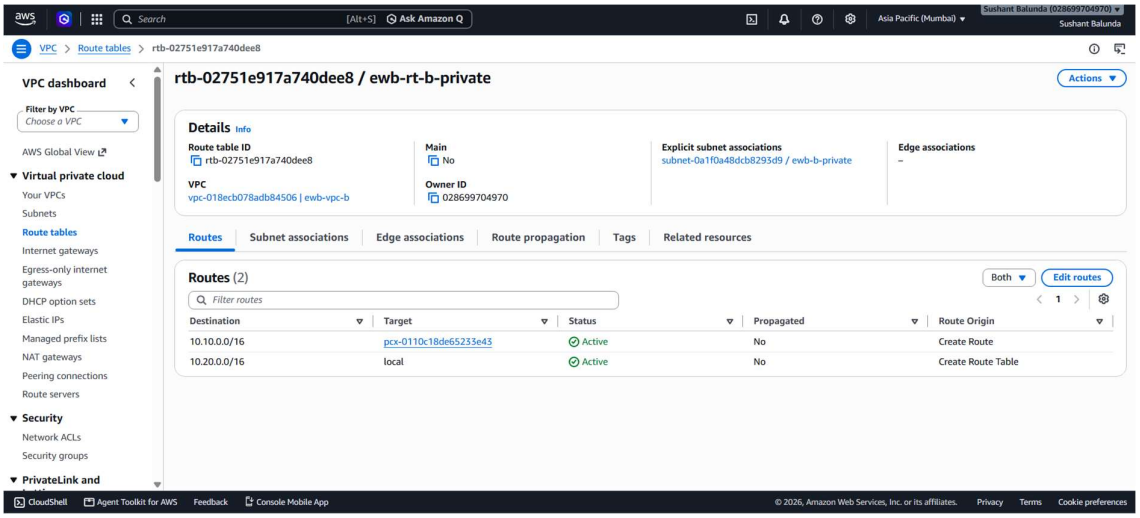


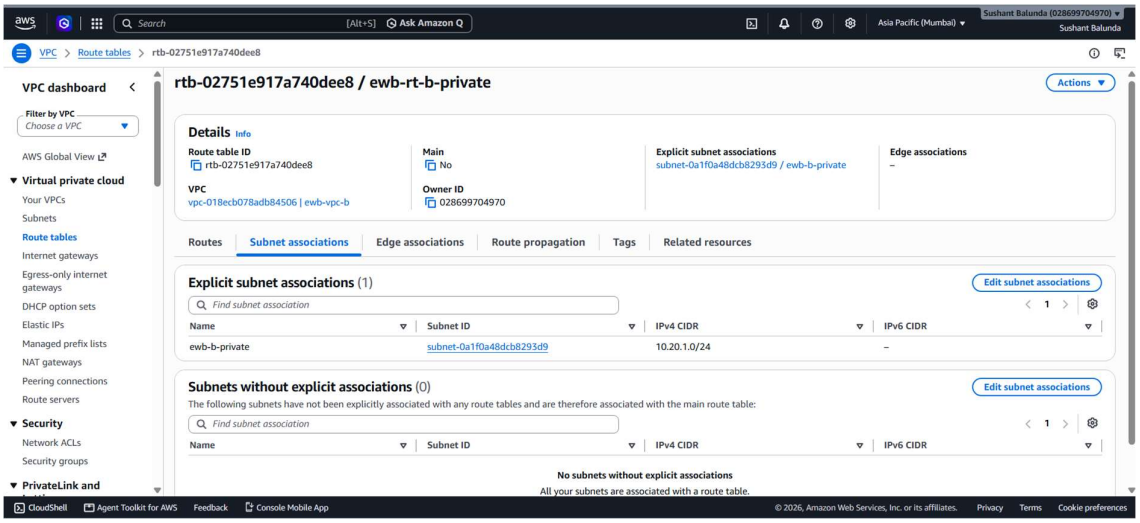



Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Route tables > rtb-0714cc0423a70f31a
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
rtb-0714cc0423a70f31a / ewb-rt-a-public
Actions
Details Info
Route table ID
rtb-0714cc0423a70f31a
Main
No
Explicit subnet associations
subnet-08bbcaf9b6d027fd3 / ewb-a-public
Edge associations
–
VPC
vpc-0e4db63a4af3e4609 | ewb-vpc-a
Owner ID
028699704970
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (1)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
ewb-a-public
subnet-08bbcaf9b6d027fd3
10.10.1.0/24
–
Subnets without explicit associations (0)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
No subnets without explicit associations
All your subnets are associated with a route table.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Route tables > rtb-026d9cc4d70680a6d
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
rtb-026d9cc4d70680a6d / ewb-rt-a-private
Actions
Details Info
Route table ID
rtb-026d9cc4d70680a6d
Main
No
Explicit subnet associations
subnet-0a3f0839ce48566e7 / ewb-a-private
Edge associations
–
VPC
vpc-0e4db63a4af3e4609 | ewb-vpc-a
Owner ID
028699704970
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
Destination
Target
Status
Propagated
Route Origin
10.10.0.0/16
local
Active
No
Create Route Table
10.20.0.0/16
pcx-0110c18de65233e43
Active
No
Create Route
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Route tables > rtb-026d9cc4d70680a6d
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
rtb-026d9cc4d70680a6d / ewb-rt-a-private
Actions
Details Info
Route table ID
rtb-026d9cc4d70680a6d
Main
No
Explicit subnet associations
subnet-0a3f0839ce48566e7 / ewb-a-private
Edge associations
–
VPC
vpc-0e4db63a4af3e4609 | ewb-vpc-a
Owner ID
028699704970
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (1)
Edit subnet associations
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
ewb-a-private
subnet-0a3f0839ce48566e7
10.10.2.0/24
–
Subnets without explicit associations (0)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
Find subnet association
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
No subnets without explicit associations
All your subnets are associated with a route table.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

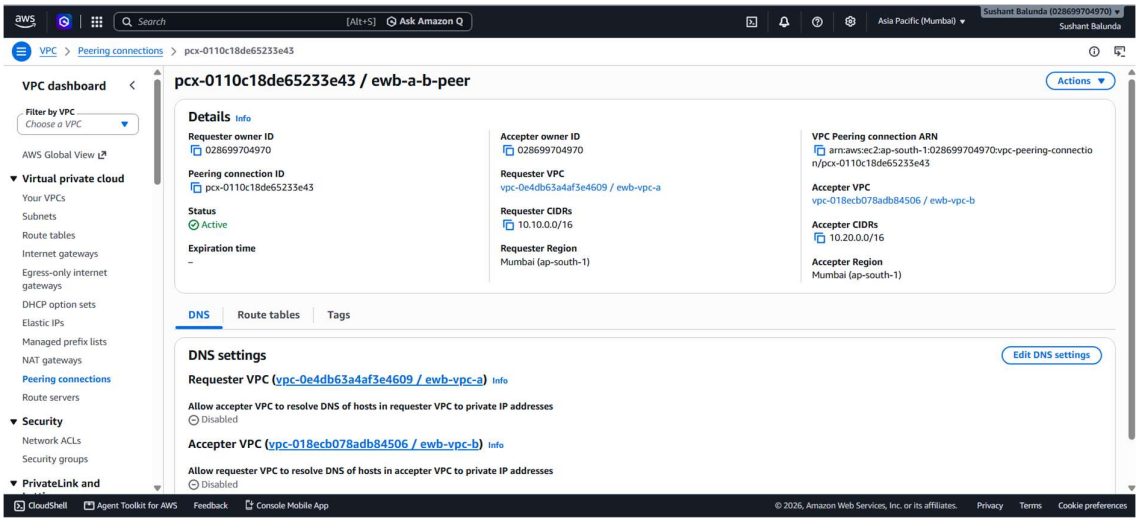

Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Route tables > rtb-02751e917a740dee8
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
rtb-02751e917a740dee8 / ewb-rt-b-private
Actions
Details Info
Route table ID
rtb-02751e917a740dee8
Main
No
Explicit subnet associations
subnet-0a1f0a48dcb8293d9 / ewb-b-private
Edge associations
–
VPC
vpc-018ecb078adb84506 | ewb-vpc-b
Owner ID
028699704970
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Routes (2)
Both
Edit routes
Filter routes
1
Destination
Target
Status
Propagated
Route Origin
10.10.0.0/16
pcx-0110c18de65233e43
Active
No
Create Route
10.20.0.0/16
local
Active
No
Create Route Table
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

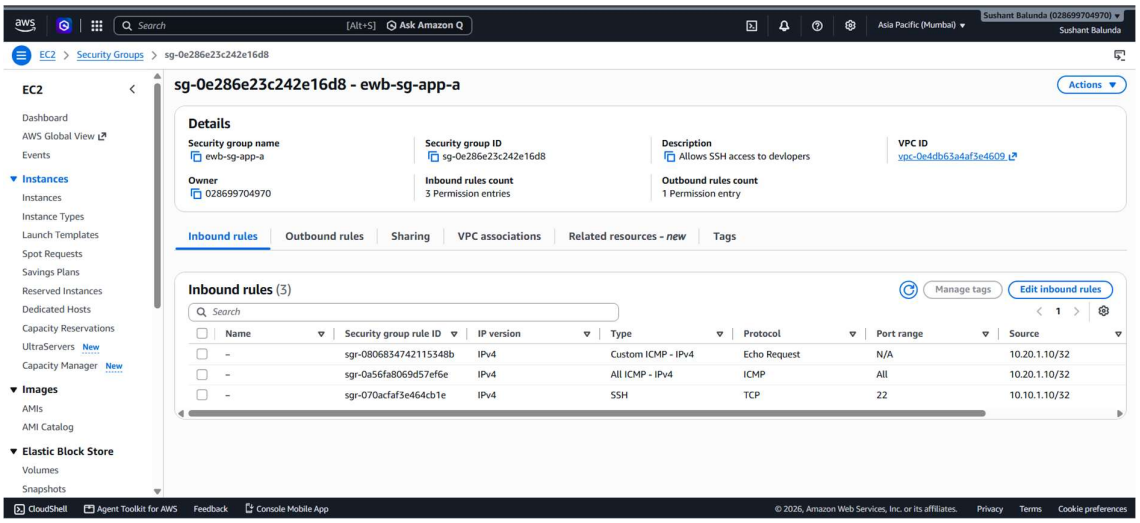

Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
VPC > Route tables > rtb-02751e917a740dee8
VPC dashboard
Filter by VPC
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and
rtb-02751e917a740dee8 / ewb-rt-b-private
Actions
Details Info
Route table ID
rtb-02751e917a740dee8
Main
No
Explicit subnet associations
subnet-0a1f0a48dcb8293d9 / ewb-b-private
Edge associations
–
VPC
vpc-018ecb078adb84506 | ewb-vpc-b
Owner ID
028699704970
Routes
Subnet associations
Edge associations
Route propagation
Tags
Related resources
Explicit subnet associations (1)
Edit subnet associations
Find subnet association
1
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
ewb-b-private
subnet-0a1f0a48dcb8293d9
10.20.1.0/24
–
Subnets without explicit associations (0)
Edit subnet associations
The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:
Find subnet association
1
Name
Subnet ID
IPv4 CIDR
IPv6 CIDR
No subnets without explicit associations
All your subnets are associated with a route table.
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

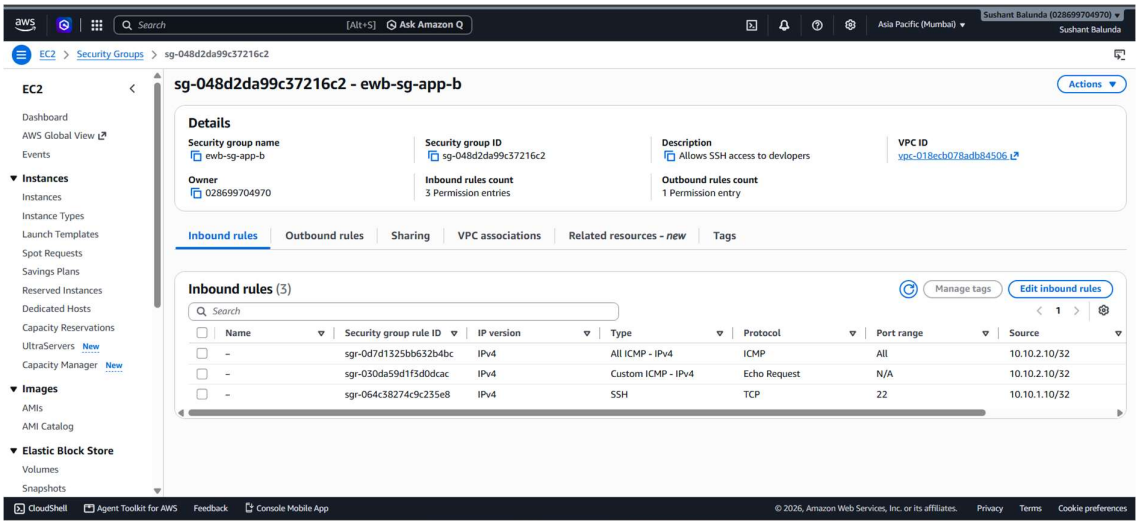
3. Active Peering and Security Groups

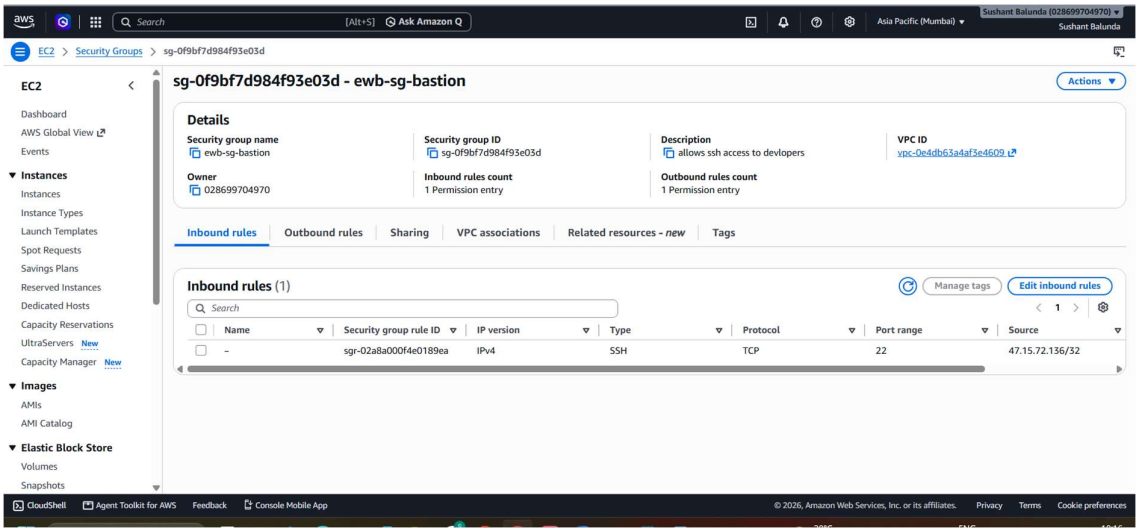
Inbound Security Rules:

- **ewb-sg-bastion:** Allows SSH (TCP 22) traffic originating from the local administrator's public IP address.
- **ewb-sg-app-a:** Allows SSH (TCP 22) traffic specifically from the bastion's private IP 10.10.1.10/32. Allows Custom ICMP - IPv4 (Echo Request) specifically from App-B's private IP 10.20.1.10/32.
- **ewb-sg-app-b:** Allows SSH (TCP 22) traffic specifically from the bastion's private IP 10.10.1.10/32. Allows Custom ICMP - IPv4 (Echo Request) specifically from App-A's private IP 10.10.2.10/32.





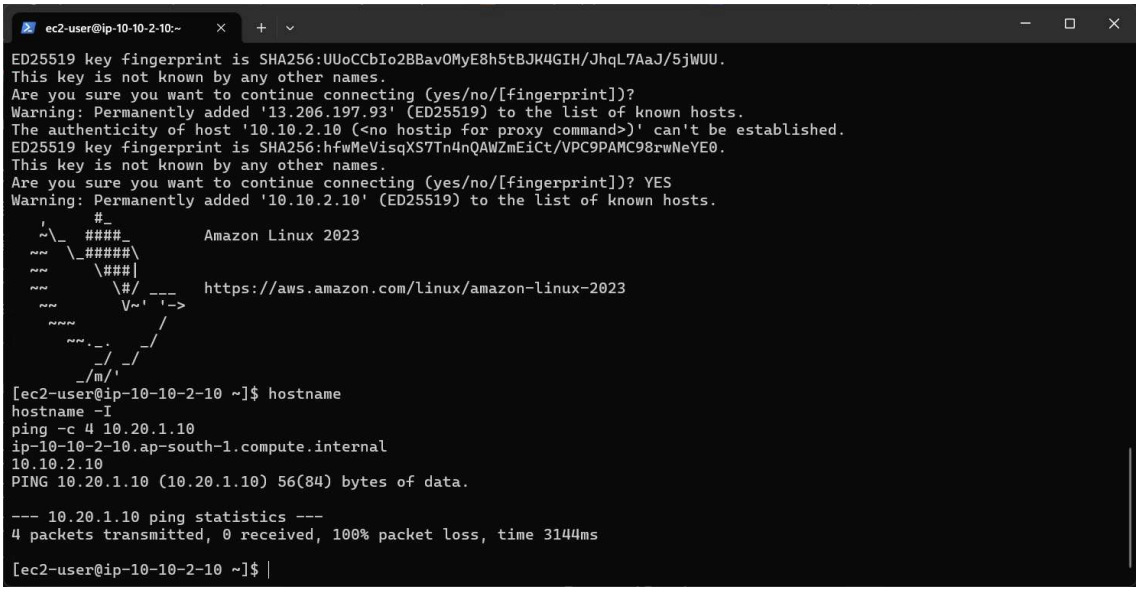

Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
EC2 > Security Groups > sg-048d2da99c37216c2
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
sg-048d2da99c37216c2 - ewb-sg-app-b
Actions
Details
Security group name
ewb-sg-app-b
Security group ID
sg-048d2da99c37216c2
Description
Allows SSH access to devlopers
VPC ID
vpc-018ecb078adb84506
Owner
028699704970
Inbound rules count
3 Permission entries
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (3)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
- sgr-0d7d1325bb632b4bc IPv4 All ICMP - IPv4 ICMP All 10.10.2.10/32
- sgr-030da59d1f3d0dcac IPv4 Custom ICMP - IPv4 Echo Request N/A 10.10.2.10/32
- sgr-064c38274c9c235e8 IPv4 SSH TCP 22 10.10.1.10/32
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search [Alt+S] Ask Amazon Q
Asia Pacific (Mumbai)
Sushant Balunda (028699704970)
Sushant Balunda
EC2 > Security Groups > sg-0f9bf7d984f93e03d
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
sg-0f9bf7d984f93e03d - ewb-sg-bastion
Actions
Details
Security group name
ewb-sg-bastion
Security group ID
sg-0f9bf7d984f93e03d
Description
allows ssh access to devlopers
VPC ID
vpc-0e4db63a4af3e4609
Owner
028699704970
Inbound rules count
1 Permission entry
Outbound rules count
1 Permission entry
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (1)
Manage tags
Edit inbound rules
Search
Name
Security group rule ID
IP version
Type
Protocol
Port range
Source
- sgr-02a8a000f4e0189ea IPv4 SSH TCP 22 47.15.72.136/32
CloudShell
Agent Toolkit for AWS
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

4. Connectivity Tests and Explanations

Connectivity Explanations:

- **Forward Path:** Traffic from ewb-app-a (10.10.2.10) destined for ewb-app-b (10.20.1.10) is directed by ewb-rt-a-private into the peering connection. The ewb-sg-app-b security group permits the incoming ICMP request.
- **Return Path:** Traffic from ewb-app-b (10.20.1.10) destined for ewb-app-a (10.10.2.10) is directed by ewb-rt-b-private back into the peering connection. The ewb-sg-app-a security group permits the incoming ICMP request.



```
[ec2-user@ip-10-10-2-10 ~]$ client_loop: send disconnect: Connection reset
client_loop: send disconnect: Unknown error
PS C:\Users\shushant>  ssh ewb-app-a
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~.-.   _/
         _/ _/
       _/m/'
Last login: Tue Oct 20 04:01:33 2026 from 10.10.1.10
[ec2-user@ip-10-10-2-10 ~]$ hostname
hostname -I
ping -c 4 10.20.1.10
ip-10-10-2-10.ap-south-1.compute.internal
10.10.2.10
PING 10.20.1.10 (10.20.1.10) 56(84) bytes of data.
64 bytes from 10.20.1.10: icmp_seq=1 ttl=127 time=0.438 ms
64 bytes from 10.20.1.10: icmp_seq=2 ttl=127 time=0.203 ms
64 bytes from 10.20.1.10: icmp_seq=3 ttl=127 time=0.198 ms
64 bytes from 10.20.1.10: icmp_seq=4 ttl=127 time=0.220 ms

--- 10.20.1.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3087ms
rtt min/avg/max/mdev = 0.198/0.264/0.438/0.100 ms
[ec2-user@ip-10-10-2-10 ~]$
```

```
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:McN20itPwn5WyZ1x6EfXw673FLjb7P6EcaA2jC38MzY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~.-.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ hostname
hostname -I
ping -c 4 10.10.2.10
ip-10-20-1-10.ap-south-1.compute.internal
10.20.1.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.193 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.204 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.210 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.196 ms

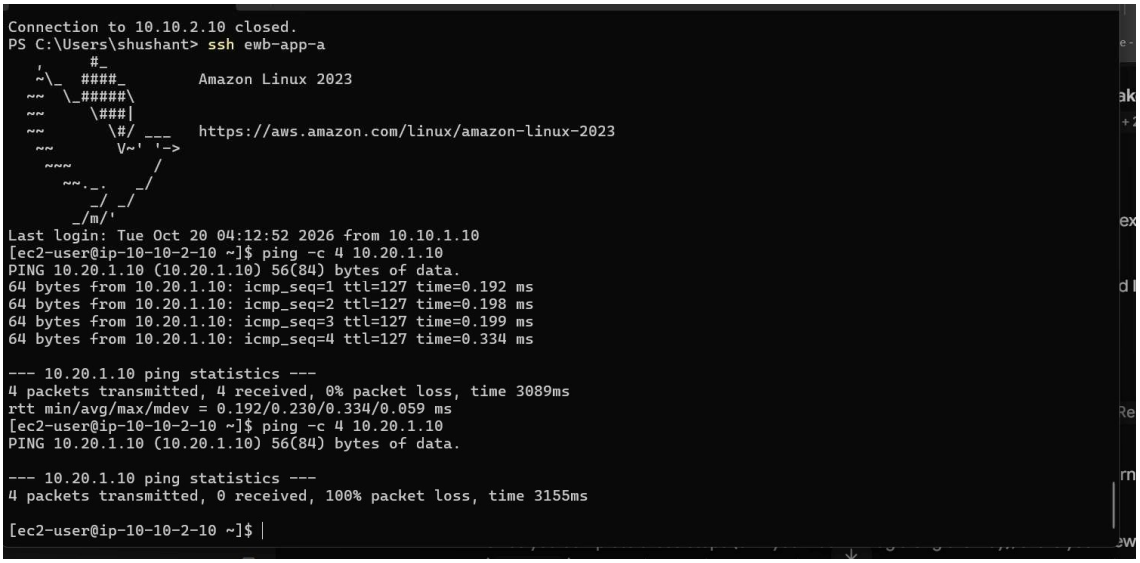
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3129ms
rtt min/avg/max/mdev = 0.193/0.200/0.210/0.006 ms
[ec2-user@ip-10-20-1-10 ~]$
```


```
The authenticity of host '10.20.1.10 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is SHA256:McN20itPwn5WyZ1x6EfXw673FLjb7P6EcaA2jC38MzY.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '10.20.1.10' (ED25519) to the list of known hosts.
   ,     #_
   ~\_  ####_        Amazon Linux 2023
  ~~  \_#####\
  ~~     \###|
  ~~       \#/ ___   https://aws.amazon.com/linux/amazon-linux-2023
   ~~       V~' '->
    ~~~         /
      ~~.-.   _/
         _/ _/
       _/m/'
[ec2-user@ip-10-20-1-10 ~]$ hostname
hostname -I
ping -c 4 10.10.2.10
ip-10-20-1-10.ap-south-1.compute.internal
10.20.1.10
PING 10.10.2.10 (10.10.2.10) 56(84) bytes of data.
64 bytes from 10.10.2.10: icmp_seq=1 ttl=127 time=0.193 ms
64 bytes from 10.10.2.10: icmp_seq=2 ttl=127 time=0.204 ms
64 bytes from 10.10.2.10: icmp_seq=3 ttl=127 time=0.210 ms
64 bytes from 10.10.2.10: icmp_seq=4 ttl=127 time=0.196 ms

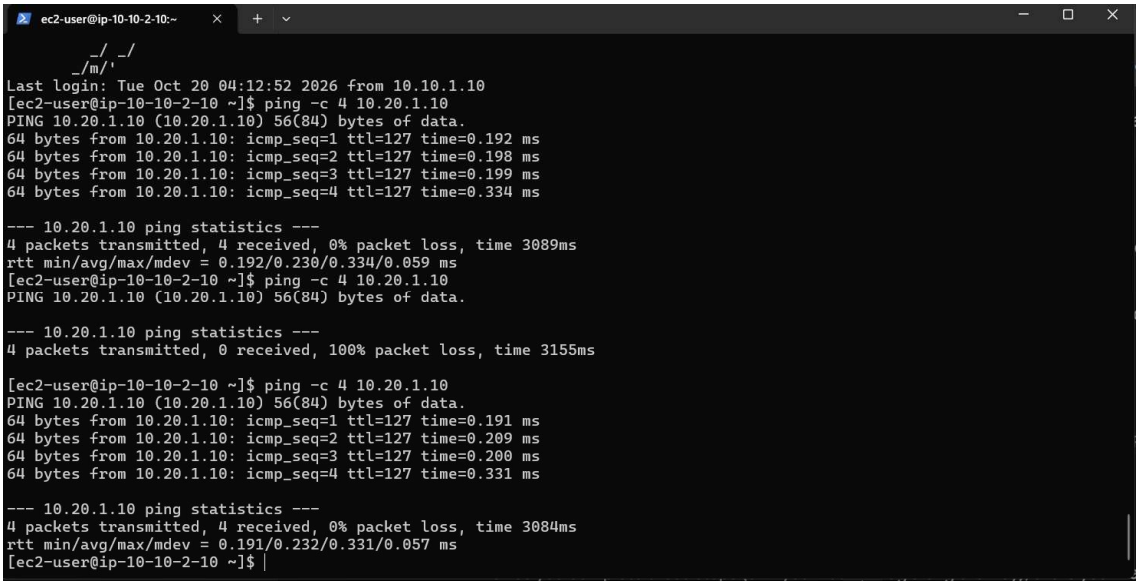
--- 10.10.2.10 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3129ms
rtt min/avg/max/mdev = 0.193/0.200/0.210/0.006 ms
[ec2-user@ip-10-20-1-10 ~]$
```

Controlled Failure & Recovery: Temporarily removing the inbound ICMP Echo Request rule from ewb-sg-app-b resulted in immediate 100% packet loss. This demonstrates that active VPC peering only provides the network path; stateful security groups must explicitly allow the traffic. Restoring the specific /32 ICMP rule immediately recovered the connection.

Controlled ICMP failure (100% packet loss after rule removal)



Controlled ICMP recovery (0% packet loss after restoring the rule)



5. Cleanup Evidence and Learnings

Cleanup Actions: All EC2 instances (ewb-bastion, ewb-app-a, ewb-app-b), the peering connection (ewb-a-b-peer), internet gateway (ewb-igw-a), custom route tables, subnets, and both VPCs (ewb-vpc-a, ewb-vpc-b) were fully deleted to prevent unwanted account charges.

Learnings: This project demonstrated that VPC peering connections are non-transitive and require precise configuration on both ends. An active peering link does not automatically guarantee connectivity; explicit return routes must be added to route tables, and stateful security groups must be tightly scoped to specific private IP addresses to establish a secure private connection.